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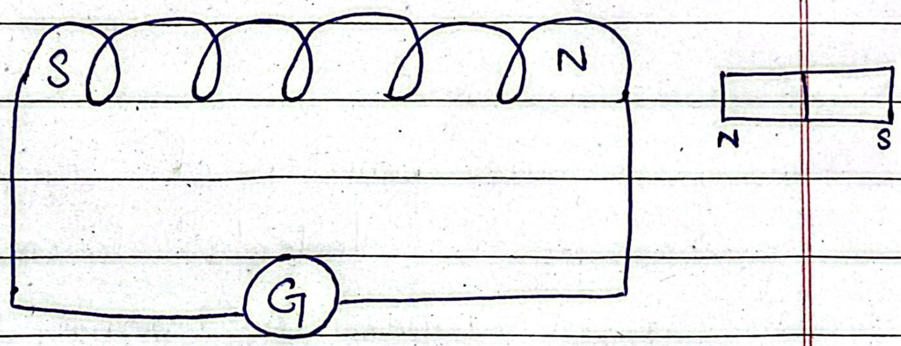
⟹ Electromagnetic Induction:

★ **Definition:**

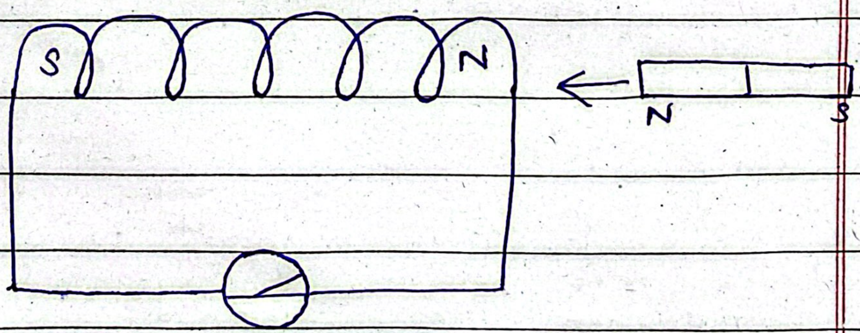
The phenomenon of generating a current in a conductor due to changing magnetic field is called electromagnetic induction.

Example:

1-



2-



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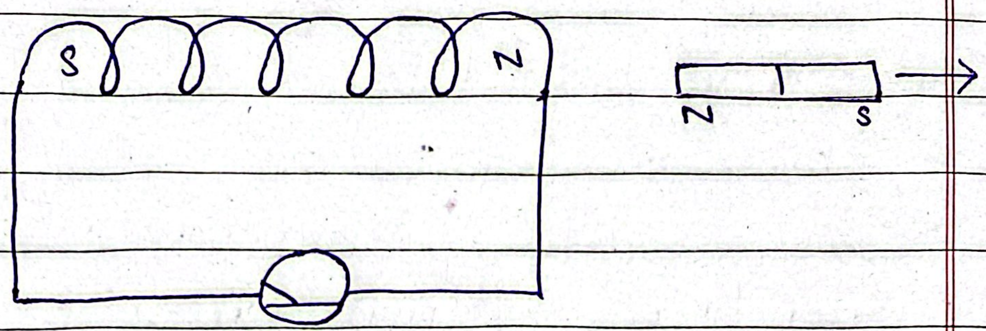
3. 

★ Case 1:

When the magnet is stationary the reading on the galvanometer is zero.

★ Case 2:

The pointer of the galvanometer deflects towards the right when the north pole of magnet is moved towards the solenoid.

★ Case 3:

If the magnet moves away from the solenoid, current flows in opposite direction. Thus, the galvanometer deflects

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towards the left.

★ **Factors:**

The factors leading to increased current in the coil are written below:

1. By using a strong magnet.
2. Increasing the motion of the magnet.
3. Increasing the area or the number of turns of the coil.

★	Position of Magnet	Deflection in Gal
1-	Magnet at rest	No deflection
2-	Magnet moves towards the coil	Deflection in galvanometer in one direction.
3-	Magnet is held stationary at same position.	No deflection